

The MathWorks Classroom Challenge Projects

Project: Maximizing Solar Panel Output for Fixed Area

Team Name: Solar Panel Team #3

Instructor: Dr. Graham

Program/Class: Engineering Pathways Program (EPP)

Members and Roles

1. Ty Schoevers - Project Manager
2. Hector Valenzuela - MATLAB Modeling Lead
3. Leo Kuraoka - Analysis & Visualization Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; crafts reports; organizes documents; ensures tasks are tracked and disagreements are resolved

- Modeling Lead: Owns core MATLAB models/simulations; codes for Energy Function and Optimization in MATLAB
- Analysis & Visualisation Lead: Verifies equations, creates plots and tests code

3) Communication Norms

- Primary channel: Discord
- Secondary channel: Email / Group Chat
- Response time: **24 hours on weekdays and 48 hours on weekends**
- Meeting cadence: One meeting per week (60–90 minutes)
- Decision records: Project Manager keeps notes in a shared Google Doc

4) Collaboration & Tools

- Version control: Shared GitHub repo / Shared Google Drive folder
- Data management: All datasets, parameters, and assumptions documented in **ReadME** or **shared Google Doc** documenting equations and MATLAB files
- File naming & style: Descriptive names using **Snake_Case**
- Backups: **CamelCase**

5) Quality Standards (Definition of 'Done')

Use MATLAB to determine the tilt angle and aspect ratio that maximize the energy output of a solar panel with a fixed area of 2 m².

Main objectives:

- Define the provided solar-energy equation in MATLAB
- Use the required boundaries
- Use MATLAB optimization tools to find the maximum energy output
- Create 3D surface plot with meshgrid and surf
- Report the optimal tilt angle, optimal aspect ratio and maximum energy

6) Timeline & Milestones

- **Week 1:** Understand the project, review MATLAB basics, define equations, assign roles, and set up the shared folder.
Milestone 1: Team agreement and role assignments completed
- **Week 2:** Create the MATLAB energy function, define optimization variables and constraints, and create the first working optimization result.
Milestone 2: Energy function coded and tested
- **Week 3:** Create the meshgrid and surf visualization, validate the optimal result, compare numerical results with the graph, and fix errors.
Milestone 3: Optimization completed
Milestone 4: 3D plot and validation completed

- **Week 4:** Complete the report, organize MATLAB files, finalize figures, prepare slides if required, practice the presentation, and submit the project

Milestone 5: Final report completed

7) Decision-Making & Conflict Resolution

- Default: Discuss as a group until **consensus is reached**
- If blocked >48 hrs: **Majority vote**
- Technical disagreements: Compare MATLAB outputs and equations before deciding.
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: Shared Google Doc checklist
- Accountability: Members communicate early if they cannot finish a task so work can be reassigned.
- Attendance: Notify the group at least 24 hours before missing a meeting whenever possible.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- Numerical value for optimal tilt angle
- Numerical value for optimal aspect ratio
- Numerical value for maximum energy output - note: this value will be in simplified arbitrary units that, in a real-world context, would more often be defined in units of solar irradiance (watts per m^2)
- Visualization of $E(\theta, r)$

12) Agreement & Signatures

Name & Signature: Ty Schoevers  Date: 7/30/2026

Name & Signature: Hector Valenzuela  Date: 07/30/2026

Name & Signature: Leo Kuraoka  Date: 8/1/2026